

The empirical recurrence for OEIS A196074 is false at every index its b-file can test, and the correct one, of order 43

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Abstract

OEIS A196074 carries an empirical recurrence of order 37 contributed by R. H. Hardin. It is false. The entry publishes 30 terms and links a b-file running to $n = 210$, so the recurrence can be tested at 163 indices and it fails at every one; the first is $n = 38$, where it exceeds the true value by 393. A transfer matrix on 1461 states — independently confirmed by direct enumeration and reproducing all 200 b-file terms — gives the recurrence that does hold. It has order 43, its first 37 coefficients are exactly the published ones, and the annihilation test certifies it from $n = 44$ onwards while returning *no threshold at all* for the published line. So the published line is the correct recurrence with its last six terms dropped.

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1 The sequence and the claim

OEIS A196074 is “Number of $n \times 4$ 0..4 arrays with each element x equal to the number its horizontal and vertical neighbors equal to 0,3,2,1,4 for $x=0,1,2,3,4$.”. It has offset 1, publishes 30 terms beginning

0, 0, 1, 8, 39, 60, 111, 343, 861, 2115, 5016, 13734, ...

and links a b-file of 200. The entry carries one formula line, an empirical recurrence of order 37:

$$\begin{aligned}
 a(n) = & a(n-2) + 4a(n-3) + 14a(n-4) + 23a(n-5) \\
 & - 2a(n-6) + 24a(n-7) + 44a(n-8) - 133a(n-9) \\
 & - 186a(n-10) - 317a(n-11) - 363a(n-12) + 372a(n-13) \\
 & + 825a(n-14) - 50a(n-15) - 513a(n-16) + 173a(n-17) \\
 & + 42a(n-18) + 441a(n-19) + 835a(n-20) + 563a(n-21) \\
 & + 336a(n-22) - 1995a(n-23) - 4044a(n-24) - 1095a(n-25) \\
 & + 4154a(n-26) + 1656a(n-27) - 395a(n-28) + 3391a(n-29) \\
 & - 709a(n-30) - 3210a(n-31) - 1608a(n-32) - 2382a(n-33) \\
 & - 47a(n-34) - 219a(n-35) + 91a(n-36) + 1032a(n-37). \quad (1)
 \end{aligned}$$

As of the live entry (revision 9, Oct 2025) this is recorded as empirical, with no range stated and nothing recording it as settled or corrected.

With offset 1 and order 37 the line first asserts something at $n = 38$. DATA stops at $n = 30$, so on the DATA alone it is untestable — but the b-file tests it 163 times.

2 The published recurrence fails at all 163 indices its b-file can test

Substituting the b-file's own integers into the right-hand side of (1):

n	$a(n)$	(1) predicts	difference
38	299231374661523	299231374661916	-393
39	748157125706655	748157125707069	-414
40	1870872919049913	1870872919049599	+314
41	4678203678514716	4678203678514596	+120
42	11697062923684722	11697062923682370	+2352
43	29247168388430532	29247168388418874	+11658

Proposition 1. (1) is false. It fails at every one of the 163 indices $n = 38, \dots, 200$ at which the entry's b-file can test it. The smallest is $n = 38$, where $a(38) = 299231374661523$ and (1) gives 299231374661916, exceeding it by 393.

Proof. Direct evaluation in exact integer arithmetic on the 200 integers the entry's b-file publishes. No model enters this computation. $\square$

3 What the entry counts, and a model for it

Write $f = (0, 3, 2, 1, 4)$, that is $f(0) = 0$, $f(1) = 3$, $f(2) = 2$, $f(3) = 1$, $f(4) = 4$. An array is an $n \times 4$ matrix over $\{0, 1, 2, 3, 4\}$, and the condition is that every cell of value x has exactly x of its horizontal and vertical neighbours equal to $f(x)$, positions outside the array not being counted. The entry states this in the same words: "Every 0 is next to 0 0's, every 1 is next to 1 3's, every 2 is next to 2 2's, every 3 is next to 3 1's, every 4 is next to 4 4's".

Because the neighbourhood reaches one row up and one row down, a state carrying a single row is not enough. Take ordered pairs of consecutive rows as vertices, with $(p, c) \rightarrow (c, x)$ an edge when every cell of the middle row c satisfies the condition with p above and x below, and mark the pairs whose first row passes with nothing above and those whose second passes with nothing below. Of the $5^8 = 390625$ ordered pairs only $S = 1461$ are reachable, and merging states with identical future behaviour leaves 58.

The model reproduces all 30 published DATA terms and all 200 b-file terms exactly. It was not taken on trust: an independent enumeration, written from the entry's wording alone and sharing no code with the model, gives

$$0, 0, 1, 8, 39, 60, 111 \quad (n = 1, \dots, 7),$$

the entry's first seven terms.

4 The recurrence that does hold

Being a walk count, a is C -finite. Solving over $\mathbf{Q}$ from the b-file's own terms, the least order that admits a recurrence is 43, and the coefficients come out integral:

$$\begin{aligned}
a(n) = & a(n-2) + 4a(n-3) + 14a(n-4) + 23a(n-5) \\
& - 2a(n-6) + 24a(n-7) + 44a(n-8) - 133a(n-9) \\
& - 186a(n-10) - 317a(n-11) - 363a(n-12) + 372a(n-13) \\
& + 825a(n-14) - 50a(n-15) - 513a(n-16) + 173a(n-17) \\
& + 42a(n-18) + 441a(n-19) + 835a(n-20) + 563a(n-21) \\
& + 336a(n-22) - 1995a(n-23) - 4044a(n-24) - 1095a(n-25) \\
& + 4154a(n-26) + 1656a(n-27) - 395a(n-28) + 3391a(n-29) \\
& - 709a(n-30) - 3210a(n-31) - 1608a(n-32) - 2382a(n-33) \\
& - 47a(n-34) - 219a(n-35) + 91a(n-36) + 1032a(n-37) \\
& 274a(n-38) + 112a(n-39) + 84a(n-40) - 56a(n-41) \\
& + 8a(n-42) - 8a(n-43). \quad (2)
\end{aligned}$$

The first 37 coefficients of (2) are, term for term, those of (1). The published line is (2) with its last six terms, $+274a(n-38) + 112a(n-39) + 84a(n-40) - 56a(n-41) + 8a(n-42) - 8a(n-43)$, missing.

Theorem 2. (2) holds for every $n > 42$.

Proof. Let $q(t) = t^{43} - \sum_i c_i t^{43-i}$ be its characteristic polynomial and $w = q(M)\mathbf{u}$, with M the transfer matrix above and $\mathbf{u}$ the start vector. For n beyond the order,

$$a(n) - \sum_i c_i a(n-i) = \mathbf{v}^\top M^{n-44} w,$$

so the recurrence holds from there on exactly when $\mathbf{v}^\top M^m w = 0$ for every $m \geq 0$; by the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so checking $m < S$ suffices. Carried out in exact integer arithmetic on the merged model, every value vanishes, and the test returns the threshold 42. $\square$

Corollary 3. (1) admits no threshold: there is no index beyond which it holds.

Proof. The same test applied to the published coefficients returns no threshold — the residuals $\mathbf{v}^\top M^m w$ for the order-37 polynomial do not all vanish, at any m within the Cayley–Hamilton bound, so they do not vanish eventually. Proposition 1 exhibits the failure at 163 consecutive indices in any case. $\square$

5 Verification

Five checks.

First, the disproof of Section 2 uses only the 200 integers of the entry's own b-file, and does not depend on the model or on Theorem 2.

Second, the model was confirmed against an independent enumeration written from the entry's wording alone, agreeing at $n = 1, \dots, 7$, and it reproduces all 30 DATA terms and all 200 b-file terms — every one, not a sample.

Third, (2) was checked directly against the b-file at all 157 indices from $n = 44$ to $n = 200$ at which it asserts anything, and holds at every one.

Fourth, every order below 43 was solved for and each fails, so 43 is minimal and the published 37 is not merely incomplete bookkeeping but too short to be repaired by any choice of coefficients.

Fifth, the annihilation test was run on both lines through the same code: it certifies (2) from $n = 44$ and returns nothing for (1).

Remark 4. This is the third entry of this shape settled here in one day, after A269637 (order 10 published, 13 true) and A236647 (34 published, 38 true). In all three the published coefficients are not wrong but incomplete, and in all three they are the true recurrence's first coefficients term for term. A recurrence fitted to a finite sample and reported at an order the sample cannot support looks exactly like this: correct as far as it goes, and false at the first index where it says anything.

References

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